

- The clusters should be more or less of equal size.
- The clusters should be more or less homogeneous in respect of the characteristics under study.

9.2.5. Multi-Stage Sampling :

Without collecting data from all the units in the selected clusters, it is possible to get relatively efficient and good estimator through random sample from each selected clusters. This method is known as two-stage sampling. At the first stage some of the clusters are selected at random and some units from each selected cluster are randomly selected at the second stage.

If the above method is extended to several stages, the sampling is termed as *multistage sampling*. For illustration, if we want to conduct a study on the farmers of a district, we may select a few upazillas at the first stage. From each of the above selected upazillas, a number of villages can be randomly selected at the second stage. From each selected village a number of farmers may be selected at random for collection of required data.

Multistage sampling is relatively more flexible. Selection of sample units and data collection is easy and administrating is more convenient. For investigation in a big area, this method is very popular.

Exercises

1. Draw a random sample (without replacement) of size 20 from a population of size 200.
2. Suppose you have a population whose elements are 3, 4, 5, 6, 7 and 8. Draw all possible samples of size 2 and prove that the mean of these sample means is equal to the mean of the population.

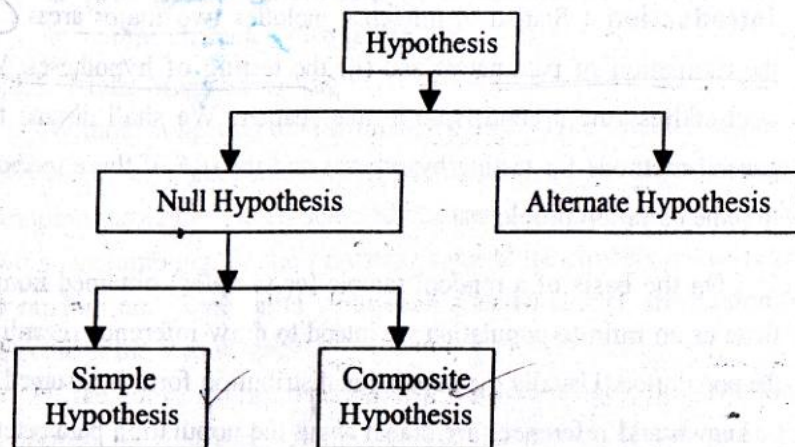
TEST OF HYPOTHESIS

Introduction : Statistical inference includes two major areas (i) the estimation of parameters and (ii) the testing of hypotheses. We shall address the second issue in this chapter. We shall discuss the general methods for testing hypotheses and the use of these methods to some common problems.

On the basis of a random sample (or samples) obtained from a finite or an infinite-population we intend to draw inference regarding the population. Usually the population distribution form is assumed to be known and inferences are drawn about the population parameters. Some hypothesis is made regarding the parameters and is tested whether it is acceptable on the basis of sample observations. For example, we may wish to compare the yield of a new crop variety with that of a standard one and perhaps recommend that the standard variety be replaced by the new one if it appears to be superior. This is an usual situation in experimental research. Before taking up a detailed discussion, it is felt necessary to explain some concepts that are of fundamental importance.

Statistical Hypothesis : Statement or assumption about population parameter(s) or about the probability distribution characterising a population (under study on the basis of sample observations) is known as a statistical hypothesis.

Hypothesis and its classification can be demonstrated as shown below :



Null hypothesis : In testing hypothesis the decision maker should maintain neutral or noncommittal attitude before sample observations are taken. Such neutrality is the keynote of the *null hypothesis* for appropriate decision making. A statistical hypothesis stated to make an appropriate decision is called a null hypothesis and usually denoted by H_0 . Any other hypothesis is called alternative hypothesis, usually denoted by H_1 or H_A .

Let $(x_1, x_2, \dots, x_n)$ be a random sample from population having a parameter θ . We wish to test the hypothesis.

$$H_0 : \theta = \theta_0.$$

Let there be some other values $\theta_1, \theta_2, \dots$ of θ which are true. Then $H_0 : \theta = \theta_0, H_1 : \theta = \theta_1, H_2 : \theta = \theta_2, \dots$ are called admissible hypotheses. The hypothesis $H_0 : \theta = \theta_0$ is the null hypothesis and every other admissible hypotheses is an alternative hypothesis.

In the example mentioned above regarding the introduction of a new variety the hypothesis that of equality of yields of both the varieties is a null hypothesis, H_0 .

A hypothesis, H_0 which specifies the values of all the parameters in the population distribution is called a simple hypothesis, otherwise the hypothesis is termed as composite hypothesis.

For example, if the population under study has two parameters μ and σ^2 , a hypothesis specifying both these parameters.

$$H_0 : \mu = \mu_0, \sigma^2 = \sigma_0^2$$

mean, variance

will be termed as a simple hypothesis.

On the other hand, if a hypothesis specifies, only one parameter, either μ or σ^2 , it is a composite hypothesis.

$$(i) H_0 : \mu = \mu_0 \text{ or}$$

$$(ii) H_0 : \sigma^2 = \sigma_0^2$$

are both composite hypothesis.

Tests of significance is usually done by two methods :

(i) Parametric statistical tests and

(ii) Non-parametric statistical tests

Parametric Test Methods : If the form of the distribution of the population under study is known beforehand and the hypothesis assumed about the population parameter(s), is known as a parametric hypothesis. As for example, the hypothesis that specifies the mean and/or the variance is a parametric hypothesis. Methods of testing parametric hypotheses are termed as parametric test methods.

Nonparametric Methods : Sometimes the precise form of the population distribution is not known and hence no assumptions are made about the precise form of the sample population. In such

situations decision are made about the population on the basis of various characteristics of the random sample. Such methods are known as nonparametric methods and the tests are categorised as nonparametric statistical tests.

Critical Region : Test statistic is used to test the null hypothesis under study. Probability distribution of such a test statistic is called sampling distribution. In order to test the null hypothesis the probability density is divided into two regions, rejection region and acceptance region. The value of the test statistic that divides the sampling distribution in rejection region and acceptance region is known as critical value (Figures 10.1 and 10.2). The rejection region is often called the critical region.

Test of Significance : In hypothesis testing we are usually concerned to make a decision regarding the null hypothesis. If a test is concerned with only the null hypothesis and not to investigate the possible alternatives and if the aim is to test whether the null hypothesis is false, the test is known as test of significance. *True, false from*

Errors in Test of Significance : Acceptance or rejection of a null hypothesis on the basis of the observations of a particular sample may not always be true for the population under study. The possible errors that we may commit in any test procedure are the following :

- (i) We may reject H_0 : when it is actually true — Type I error
- (ii) We may accept H_0 : when it is actually false.

The first one is called the type I error or rejection error. The probability of commitment of the type I error is called the level of significance, usually denoted by α . It is also known as the size of the test.

The second one is known as the acceptance error or the type II error. The probability of type II error is denoted by β . $1-\beta$ is known as the power of a test. *1-β is power of Test*

One Tailed Test : A null hypothesis usually states the equality of the parameter and the test statistic. An alternative hypothesis may assume the parameter either smaller or larger than the test statistic. Such a test which assumes an one-directional alternative hypothesis is known as an one-tailed test.

For $H_0 : \theta = \theta_0$, *equality*

$H_1 : \theta > \theta_0$ or $H_1 : \theta < \theta_0$ will be the possible alternatives, which indicates one tailed test.

Two-Tailed Test : If the alternative hypothesis states the inequality of the parameter and the test statistic, the test is called a 2-tailed test. In such case the alternative to the null hypothesis $H_0 : \theta = \theta_0$ will be $H_1 : \theta \neq \theta_0$. *not equality*

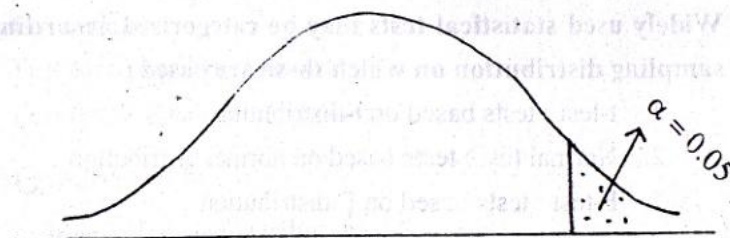


Figure 10.1 : One-tailed test at 5 % level of significance.

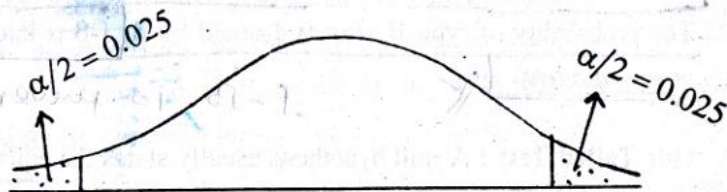


Figure 10.2 : Two-tailed test at 5 % level of significance.

Degrees of Freedom : Test of significance of population parameter is done on the basis of random sample drawn from the population. Sample size is, therefore, very important and is considered through the degrees of freedom. The number of independent observations is termed as the degrees of freedom. The number of degrees of freedom, in general, is the total number of independent observations. As a simple example, let us consider a sample of 10 observations and the mean of the 10 observations is 25. Nine of these 10 observations may have different values, but the size of the tenth observation is determined by those nine, because the mean is 25 in any case. In each case we have $10 - 1 = 9$ degrees of freedom. $df = n - 1$

Widely used statistical tests may be categorized according to the sampling distribution on which these are based :

1. t-test : tests based on t-distribution
2. Normal test : tests based on normal distribution
3. F-test : tests based on F-distribution
4. χ^2 -tests : tests based on χ^2 -distribution.

(Normal distribution has been briefly discussed in chapter 7 but t, F and χ^2 distributions are beyond the scope of this book).

We shall, however, discuss the important statistical tests according to parameter addressed. It may be mentioned that these

tests are based on a common assumption that the population under investigation has a normal distribution.

In this chapter we shall limit our discussion within the following parametric tests :

1. Tests for mean

- (i) Test for a population mean
- (ii) Test for two population means
- (iii) Test for several population means.

2. Tests for population proportion

- (i) Test for a population proportion
- (ii) Test for two population proportions.

3. Tests for regression and correlation coefficients

- (i) Test for an observed correlation coefficient
- (ii) Test for an observed rank correlation coefficient
- (iii) Test for two correlation coefficients
- (iv) Test for an observed regression coefficient
- (v) Test for two regression coefficients.

4. Tests for variance

- (i) Test for a population variance
- (ii) Test for two population variances.

5. Test of goodness of fit

6. Test of independence of attributes.

10.1. Tests for Population Mean :

- (i) Testing the significance of difference between an assumed population mean μ_0 and a sample mean $\bar{x}$.

Let us consider a random sample $(x_1, x_2, \dots, x_n)$ with mean $\bar{x}$ obtained from a normal population having mean μ .

We wish to test the null hypothesis $H_0: \mu - \mu_0 = 0$.

(The hypothesis is usually stated as $H_0: \mu = \mu_0$).

Case-1 : If the population variance σ^2 is unknown and the sample is small ($n < 30$), the appropriate test is a Student's-t test.

The test statistic is $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ with $(n-1)$ d.f.

where $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$ is the sample variance.

The test statistic t will be compared with student's t -distribution.

The test may be either one tailed or two-tailed.

There is an alternative way of testing the null hypothesis. That is by constructing the confidence interval and observing whether the interval contains the assumed mean μ_0 . If it does not contain μ_0 , the null hypothesis is rejected. The $\alpha\%$ confidence interval for μ is

$$\bar{x} \pm t_{\alpha} \times \frac{s}{\sqrt{n}}$$

Example 10.1 :

Per hectare yield of an HYV rice in 10 randomly selected agricultural substations are

63, 63, 66, 70, 70, 67, 71, 68, 69 and 71 quintals.

On the basis of the above data could we consider that the average yield of the rice variety is 66 quintals?

Here $H_0: \mu = 66$ and $H_1: \mu \neq 66$.

Sample mean is $\bar{x} = 67.8$.

An estimate of the unknown population standard deviation is

$$s = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2} = 3.011$$

The test statistic $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{67.8 - 66}{3.011/\sqrt{10}} = 1.89$ (app); with $(10-1=9)$

d.f.

The critical value of t at 5% level of significance with 9 d.f. is 2.26, the calculated value is less than the critical value.

Hence the null hypothesis may be accepted.

Alternative Procedure :

5% confidence interval for mean is

$$\bar{x} \pm t_{0.05,9} s/\sqrt{n} = 67.8 \pm 2.26 \times 0.952 = 67.8 \pm 2.152$$

Here the interval includes the given mean $\mu = 66$.

Hence we have no reason to reject the null hypothesis.

Case-2: If the population variance is unknown but the sample is large ($n \geq 30$), the appropriate test is normal test and the test

statistic is $z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ is distributed as $N(0, 1)$ distribution.

The value of z is compared with the standardized normal distribution using either a one tailed test or a two-tailed test.

Example 10.2 :

Suppose on the basis of a random sample of size 50

We have $\bar{x} = 10$ and $s = 3$

We wish to test $H_0: \mu = 7.5$ against $H_1: \mu \neq 7.5$

The test statistic

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{10 - 7.5}{3/\sqrt{50}} = 5.893$$

5% and 1% critical values of z are 1.96 and 2.58 respectively.

The null hypothesis is rejected.

Case-3: If the population variance σ^2 is known, the test statistic z will be approximately distributed as $N(0, 1)$ distribution; irrespective of sample size (small or large).

$$\text{The test statistic is } z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

is compared with the standardized normal distribution using either one-tailed or two-tailed test.

Example 10.3:

Given that $n=9$, $\bar{x} = 4.6$ and $\sigma = 1.0$

We wish to test $H_0: \mu = 4.0$ against $H_1: \mu \neq 4.0$

$$\text{The test statistic, } z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{4.6 - 4.0}{1.0 / \sqrt{9}} = 1.8$$

Which is less than the 5% critical value of z .

Hence, H_0 may be accepted.

(ii) **Testing the significance of the difference between the mean of two normal populations on the basis of independent random samples.**

Case-1: Population variances are unknown but equal; samples are small ($n_1, n_2 < 30$).

The testing hypothesis is $H_0: \mu_1 - \mu_2 = 0$ (usually stated as $H_0: \mu_1 = \mu_2$)

Let $(x_{11}, x_{12}, \dots, x_{1n_1})$ and $(x_{21}, x_{22}, \dots, x_{2n_2})$ be two independent random samples of size n_1 and n_2 from normal populations having equal but unknown variance, and unknown means μ_1 and μ_2 .

The test statistic is

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \text{ with } (n_1 + n_2 - 2) \text{ d.f.}$$

$$= \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad (\text{since } \mu_1 = \mu_2 \text{ under } H_0)$$

$$\text{where } s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}; s_1^2 = \frac{1}{n_1 - 1} \sum (x_{1i} - \bar{x}_1)^2 \text{ and } s_2^2 = \frac{1}{n_2 - 1} \sum (x_{2j} - \bar{x}_2)^2$$

The test statistic may be compared with Student's t distribution with $(n_1 + n_2 - 2)$ degrees of freedom. The test may be either one tailed or two-tailed.

Example 10.4 :

Two levels of a certain fertilizer are tried to compare their effects on the increase in yield of a local rice variety. The following are two random samples of observations.

Per hectare yield in quintals

Level 1 (x_1) = 34, 32, 30, 35, 34, 34, 32, 30, 31, 35, 35

Level 2 (x_2) = 22, 24, 22, 20, 26, 21, 20, 20, 22, 25, 26, 21, 25

We wish to test that there is no difference in the average effects of the two levels of the fertilizer.

That is, $H_0: \mu_1 = \mu_2$

Here $n_1 = 11$ $\bar{x}_1 = 32.91$ $s_1^2 = 3.89$

$n_2 = 13$ $\bar{x}_2 = 22.62$ $s_2^2 = 5.26$

$$\text{Now } s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \frac{10 \times 3.89 + 12 \times 5.26}{22} = 2.15$$

The test statistic $t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{32.91 - 22.62}{2.15 \sqrt{\frac{1}{11} + \frac{1}{13}}} = 11.96$

1% critical value of t with 22 d.f. is 2.82

Hence the null hypothesis is rejected.

Case 2: Population variances are equal but unknown; samples are large ($n_1, n_2 \geq 30$). The null hypothesis is $H_0: \mu_1 = \mu_2$

The test statistic, $z = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ is approximately

distributed as a standardised normal variate.

where $s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$

$s_1^2 = \frac{1}{n_1 - 1} \sum (x_{1i} - \bar{x}_1)^2$ and $s_2^2 = \frac{1}{n_2 - 1} \sum (x_{2j} - \bar{x}_2)^2$

Example 10.5 :

Suppose two independent random samples representing the yield of two varieties of HYV rice give the following results :

$n_1 = 30 \quad \bar{x}_1 = 12.5 \quad s_1^2 = 4.55$

$n_2 = 35 \quad \bar{x}_2 = 10.3 \quad s_2^2 = 6.25$

Now, $s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = 5.47 \quad \therefore s = 2.34$

For testing $H_0: \mu_1 = \mu_2$

The test statistic is $z = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{12.5 - 10.3}{2.34 \sqrt{\frac{1}{30} + \frac{1}{35}}} = 3.78$

As 1% critical value of z is 2.58, the null hypothesis is rejected.

Case 3: Population variances are unequal and unknown, samples are small ($n_1, n_2 < 30$) :

For testing $H_0: \mu_1 = \mu_2$, we briefly state a suggestion by Cochran and Cox (1957). They considered the statistic t' given by

$$t' = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

This t' , is not a student's t . The theoretical value of t' for judging the significance of the test statistic is as follows :

$$t'_\alpha = \frac{\frac{s_1^2}{n_1} t_1 + \frac{s_2^2}{n_2} t_2}{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Where t_1 and t_2 are students' t with $(n_1 - 1)$ and $(n_2 - 1)$ d.f. respectively at α level of significance. We reject H_0 if $t' > t'_\alpha$

Example 10.6 :

The following are the yields in quintals per hectare of two rice varieties (Irri and Boro) from 5 randomly chosen plots each. Assuming unequal population variances, we wish to test whether the two varieties of rice differ in yields.

Irri (x_1): 54 59 47 66 44

Boro (x_2): 40 45 35 44 36

Here $n_1 = 5, \bar{x}_1 = 54 \quad s_1^2 = 79.5$

$n_2 = 5, \bar{x}_2 = 40 \quad s_2^2 = 20.5$

For testing $H_0: \mu_1 = \mu_2$, the test statistic is

$$t' = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{54 - 40}{\sqrt{\frac{79.5}{5} + \frac{20.5}{5}}} = 3.13$$

The theoretical value of t' at 5% level of significance is

$$t'_{0.05} = \frac{\frac{s_1^2}{n_1} t_1 + \frac{s_2^2}{n_2} t_2}{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \frac{15.9 \times 2.132 + 4.1 \times 2.132}{20}$$

= 2.132 for one tailed test.

Here $t' > t'_{0.05}$. Hence H_0 is rejected.

Case-4: Population variances are unequal and unknown, samples are large ($n_1, n_2 \geq 30$):

For testing $H_0: \mu_1 = \mu_2$ the test statistic

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

follows standardized normal distribution.

Example 10.7:

The per hectare yield of wheat estimated on the basis of 50 randomly selected plots in one region is 30.5 quintals and standard deviation 3.5. 65 randomly chosen plots in another region gives the yield 28.0 quintals and the standard deviation 2.5. We wish to test whether the average yield of wheat in the two regions differs significantly.

The null hypothesis is $H_0: \mu_1 = \mu_2$

Given that

$n_1 = 50$	$\bar{x}_1 = 30.5$	$s_1 = 3.5$
$n_2 = 65$	$\bar{x}_2 = 28.0$	$s_2 = 2.5$

$$\text{The test statistic is } z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{30.5 - 28.0}{\sqrt{\frac{(3.5)^2}{50} + \frac{(2.5)^2}{65}}} = 4.28$$

which is greater than the critical value at 1% level of significance.

Hence H_0 is rejected.

Case-5: Population variances are equal and known

Let the population variances $\sigma_1^2 = \sigma_2^2 = \sigma^2$

For testing $H_0: \mu_1 = \mu_2$,

$$\text{The test statistic is } z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

is a standardized normal variate.

Example 10.8:

Two independent random samples from normal populations having equal and known variance give the following data.

$n_1 = 9$, $\bar{x}_1 = 1.2$, $n_2 = 16$, $\bar{x}_2 = 1.7$, $\sigma = 1.4405$

We want to test,

$H_0: \mu_1 = \mu_2$

$$\text{Our test statistic is } z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{1.2 - 1.7}{1.4405 \sqrt{\frac{1}{9} + \frac{1}{16}}} = -0.833$$

The 5% critical value of z is ± 1.96 .

Hence the null hypothesis may be accepted.

Case-6: Population variances are known but unequal

If the population variances $\sigma_1^2 \neq \sigma_2^2$ are both known, we want to test

$H_0: \mu_1 = \mu_2$ using the test statistic

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

which is distributed as a standardized normal distribution.

Example 10.9 :

The following data are recorded from two independent random samples from normal populations; population variances are unequal but known :

$$n_1 = 13 \quad \bar{x}_1 = 80.02 \quad \sigma_1^2 = 0.000576$$

$$n_2 = 8 \quad \bar{x}_2 = 79.98 \quad \sigma_2^2 = 0.001089$$

We wish the $H_0: \mu_1 = \mu_2$

$$\text{The test statistic is } z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{80.02 - 79.98}{\sqrt{\frac{0.000576}{13} + \frac{0.001089}{8}}} = 2.98$$

Critical values of z at 5% and 1% levels of significance are 1.95 and 2.58 respectively.

Hence the null hypothesis is rejected.

(iii) Testing the significance of the difference between the means of two populations on the basis of two correlated random samples (Paired t-test)

Let there be n pairs of observations and the difference between the observations of the i th pair be d_i . That is,

$$d_i = x_{1i} - x_{2i}; (i = 1, 2, \dots, n)$$

The sample mean of d_i is $\bar{d} = \bar{x}_1 - \bar{x}_2$

For testing the null hypothesis $H_0: \mu_1 = \mu_2$, the test statistic is

$$t = \frac{\bar{d}}{s_d / \sqrt{n}} \text{ with } (n-1) \text{ d.f.}$$

$$\begin{aligned} \text{where, } s_d^2 &= \frac{1}{n-1} \sum (d_i - \bar{d})^2 = s_1^2 + s_2^2 - \frac{2\sum (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)}{n-1} \\ &= s_1^2 + s_2^2 - 2 \text{cov.}(x_1, x_2) = s_1^2 + s_2^2 - 2r s_1 s_2 \end{aligned}$$

where s_1^2 and s_2^2 are the sample variances.

The test may be either one-tailed or two-tailed.

Example 10.10:

Six experimental units (plots of land) are each divided in to two equal parts and two types of manures are applied to compare their effects on rice yield. Per plot yield in kg. are given below :

Manure	Yield					
Manure-1	20.50	24.60	23.06	23.98	25.50	22.07
Manure-2	24.86	26.39	28.19	28.75	30.35	24.05

In order to test the null hypothesis $H_0: \mu_1 = \mu_2$

The pair differences d_i are as shown below :

Yield of manure-1 (x_1)	Yield of manure-2 (x_2)	$d_i = x_{2i} - x_{1i}$
20.50	24.86	4.36
24.60	26.39	1.79
23.06	28.19	5.13
23.98	28.75	4.77
25.50	30.35	4.85
22.07	24.05	1.98

We compute $\bar{d} = 3.813$ $s_d = 1.5151$

$$\text{Test statistic } t = \frac{\bar{d}}{s_d / \sqrt{n}} = \frac{3.813}{1.5151 / \sqrt{6}} = 6.165$$

For $6-1=5$ degrees of freedom the 5% and 1% critical values of t are 2.571 and 4.032 respectively.

Hence the null hypothesis is rejected!

10.2. Tests for Population Proportion :

- (i) Testing the significance of the difference between an assumed proportion p_0 and an observed proportion p .

The null hypothesis is $H_0 : p = p_0$

Let $(x_1, x_2, \dots, x_n)$ be a random sample from a population in which it is assumed that a proportion p_0 belongs to a specified class. The proportion p of elements in the sample belonging to this class is calculated. The test statistic is

$$z = \frac{p - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

This may be compared with a standardized normal distribution using either a one-tailed or a two-tailed test.

It is to be noted that the test is approximate and assumes that the sample size is sufficiently large ($n \geq 30$) to justify normal approximation of binomial.

Example 10.11:

Suppose a random sample of size 100 gives an observed proportion of 0.4. We wish to test if any item to be included in the class under consideration has fifty percent possibility. That is, we wish to test $H_0 : p = 0.5$.

The test statistic

$$z = \frac{p - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.4 - 0.5}{\sqrt{\frac{0.5(1-0.5)}{100}}} = -2.0$$

Critical value $z_{(0.05)} = \pm 1.96$, the calculated value falls outside the critical region and hence H_0 is rejected.

(ii) Testing the Significance of Difference Between Two Population Proportions

Let the populations have proportions π_1 and π_2 with the same characteristic. Random samples of size n_1 and n_2 are taken and respective proportions p_1 and p_2 are calculated. For testing

$$H_0 : \pi_1 = \pi_2$$

$$\text{The test statistic is } z = \frac{p_1 - p_2}{\sqrt{p(1-p) \left\{ \frac{1}{n_1} + \frac{1}{n_2} \right\}}}$$

$$\text{where, } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

z is approximately distributed as a standardized normal variate and the resulting test may be either one-tailed or two-tailed.

The test is approximate and the samples should be large enough ($n_1, n_2 \geq 30$) to justify the normal approximation of binomial.

Example 10.12:

Two batches of jute seeds were tested for germination capacity. two independent random samples, one from each batch gave the following results :

1st Sample : $n_1 = 952$ Proportion of non-germinating seeds being 0.0325

2nd Sample : $n_2 = 1168$ Proportion of non-germinating seeds being 0.0573

We wish to test whether the proportions of non-germinating seeds in the two batches differ significantly.

Here the null hypothesis is $H_0 : \pi_1 = \pi_2$

We have, $n_1 = 952$, $p_1 = 0.0325$ and $n_2 = 1168$, $p_2 = 0.0573$

$$\text{and } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = 0.0467$$

The test statistic

$$z = \frac{p_1 - p_2}{\sqrt{p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{-0.0248}{\sqrt{(0.0462)(0.9538)\left(\frac{1}{952} + \frac{1}{1168}\right)}} = -2.71^*$$

As the 5% critical value is ± 1.96 , the calculated value falls beyond the critical region and hence H_0 is rejected.

10.3. Tests for Correlation Coefficients:

(i) Testing the significance of an observed correlation coefficient.

On the basis of a random sample $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ from a bivariate normal population we have to test the hypothesis that the population correlation coefficient is statistically significant.

The null hypothesis is $H_0: \rho = 0$

The test statistic is $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$ with $(n-2)$ d.f.

Where r is the sample correlation coefficient.

Example 10.13: A random sample of chicks in a poultry farm are given two different feeds in alternate weeks (keeping a suitable gap period) and weight gain (%) are recorded as shown below:

Feed 1 (x): 21 25 26 24 22 30 19 24 28 32 31 29 21 18

Feed 2 (y): 19 20 24 21 21 24 18 12 19 30 27 26 19 18

We wish to test the significance of the coefficient of correlation between the feed effects.

First we compute the sample correlation coefficient r as

$$r = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\left\{\sum x^2 - \frac{(\sum x)^2}{n}\right\} \left\{\sum y^2 - \frac{(\sum y)^2}{n}\right\}}} = \frac{195}{\sqrt{264 \times 270.86}} = 0.73$$

For testing the null hypothesis $H_0: \rho = 0$ we compute the test

$$\text{statistic } t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.73\sqrt{14-2}}{\sqrt{1-(0.73)^2}} = 3.70$$

which is greater than the theoretical value 3.06 of t at 1% level of significance (12 d.f.).

Hence the null hypothesis is rejected and the correlation between the feed effects is considered as highly significant.

(ii) Testing the Significance of an Observed Rank Correlation Coefficient.

The null hypothesis is, $H_0: \rho_k = 0$ or, $\rho_s = 0$

(ρ_k and ρ_s being the Kendall's and Spearman's rank correlation coefficient respectively).

$$\text{The test statistic is, } t = \frac{r_k \sqrt{n-2}}{\sqrt{1-r_k^2}} \text{ or, } t = \frac{r_s \sqrt{n-2}}{\sqrt{1-r_s^2}} \text{ with } (n-2) \text{ d.f.}$$

Where r_k and r_s are the Kendall's and Spearman's rank correlation coefficient for the sample.

(iii) Testing the Significance of Difference Between a Correlation Coefficient and an Assumed Value ρ_0 .

The null hypothesis is, $H_0: \rho = \rho_0$

With sample correlation coefficient r using z transformation we have

$$z_1 = \frac{1}{2} \log_e \left(\frac{1+r}{1-r} \right) = \frac{1}{2} \times 2.3026 \log_{10} \left(\frac{1+r}{1-r} \right)$$

The distribution of z_1 is approximately normal with mean μ_{z_1} and standard deviation σ_{z_1} were

$$\mu_{z_1} = \frac{1}{2} \log_e \left(\frac{1+\rho_0}{1-\rho_0} \right) = \frac{1}{2} \times 2.3026 \log_{10} \left(\frac{1+\rho_0}{1-\rho_0} \right)$$

$$\text{and } \sigma_{z_1} = \frac{1}{\sqrt{n-3}}$$

The test statistic becomes

$$z = \frac{z_1 - \mu_{z_1}}{\sigma_{z_1}} \text{ which is a standardized normal variate.}$$

Example 10.14:

On the basis of sample correlation coefficient $r = 0.75$ between the variables of the random sample of size 24 from a bivariate normal population we wish to test the null hypothesis.

$$H_0: \rho = 0.50$$

$$\text{Here } \mu_{z_1} = 1.1513 \log_{10} \left(\frac{1+0.50}{1-0.50} \right) = 0.5493$$

$$z_1 = 1.1513 \log_{10} \left(\frac{1+0.75}{1-0.75} \right) = 0.9730$$

$$\sigma_{z_1} = \frac{1}{\sqrt{n-3}} = \frac{1}{\sqrt{24-3}} = 0.2182$$

The test statistic

$$z = \frac{z_1 - \mu_{z_1}}{\sigma_{z_1}} = \frac{0.97730 - 0.5493}{0.2182} = 1.94$$

The critical value at $\alpha = 0.05$ is 1.96

The calculated value is smaller than the critical value.

Hence the null hypothesis may be accepted.

(iv) Testing the Significance of the Difference Between Two Population Correlation Coefficients:

Let us consider two random samples of size n_1 and n_2 from two bivariate normal populations having correlation coefficients ρ_1 and ρ_2

respectively. Let the sample correlation coefficients be r_1 and r_2 respectively.

We wish to test the null hypothesis

$$H_0: \rho_1 = \rho_2$$

$$\text{We first compute } z_1 = \frac{1}{2} \log_e \left(\frac{1+r_1}{1-r_1} \right) \text{ and } z_2 = \frac{1}{2} \log_e \left(\frac{1+r_2}{1-r_2} \right)$$

Under H_0 , $z_1 - z_2$ will follow normal distribution with mean zero and variance $\left(\frac{1}{n_1-3} + \frac{1}{n_2-3} \right)$.

Hence the test statistic becomes

$$z = \frac{z_1 - z_2}{\sqrt{\frac{1}{n_1-3} + \frac{1}{n_2-3}}}$$

which follows standardized normal distribution.

Example 10.15:

Given that, $n_1 = 28$ $n_2 = 35$ $r_1 = 0.50$ and $r_2 = 0.30$

$$\text{Now } z_1 = 1.1513 \log_{10} \left(\frac{1+0.50}{1-0.50} \right) = 0.5493$$

$$z_2 = 1.1513 \log_{10} \left(\frac{1+0.30}{1-0.30} \right) = 0.3095$$

$$\sigma = \sqrt{\frac{1}{28-3} + \frac{1}{35-3}} = 0.2669$$

$$\text{The test statistic, } z = \frac{0.5493 - 0.3095}{0.2669} = 0.90$$

Which is smaller than the 5% critical value (1.96).

Hence the null hypothesis $H_0: \rho_1 = \rho_2$ may be accepted.

10.4. Tests for Regression Coefficients:

(i) Testing the Significance of an Observed Regression Coefficient:

A random sample $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ is drawn from a bivariate normal population. We wish to test the hypothesis if the regression of y on x is statistically significant. The null hypothesis may be (a) $H_0: \beta = 0$, or (b) $H_0: \beta = \beta_0$.

(a) For testing $H_0: \beta = 0$, we first compute the sample regression coefficient b as usual.

The test statistic is $t = \frac{b}{s_b}$ with $(n-2)$ d.f.

Where variance of b is obtained as $s_b^2 = \frac{S^2}{SS(x)}$

$$\text{Now } S^2 = \frac{1}{n-2} \left[SS(y) - \frac{\{SP(xy)\}^2}{SS(x)} \right] = \frac{SS(x)SS(y) - \{SP(xy)\}^2}{(n-2)SS(x)}$$

$$\Rightarrow t = \frac{bSS(x) \sqrt{n-2}}{\sqrt{SS(x)SS(y) - \{SP(xy)\}^2}} = \frac{SP(xy) \sqrt{n-2}}{\sqrt{SS(x)SS(y) - \{SP(xy)\}^2}}$$

The test may be two-tailed since b may be positive or negative.

Example 10.16:

A random sample of size 12 gives the following results:

$$\Sigma x_i = 766 \quad \Sigma y_i = 1700 \quad \Sigma x_i^2 = 49068 \quad \Sigma y_i^2 = 246100$$

$$\text{and } \Sigma x_i y_i = 109380$$

$$\text{We compute } SS(x) = 171.67 \quad SS(y) = 5266.67$$

$$SP(xy) = 863.33$$

$$\text{and } b = \frac{SP(xy)}{SS(x)} = 5.029$$

$$\begin{aligned} \text{Now } t &= \frac{SP(xy) \sqrt{n-2}}{\sqrt{SS(x)SS(y) - \{SP(xy)\}^2}} \\ &= \frac{863.33 \sqrt{10}}{\sqrt{171.67 \times 5266.67 - (863.33)^2}} = 6.85 \end{aligned}$$

The 5% and 1% critical value of t with 10 d.f. are 2.228 and 3.169 respectively. The null hypothesis is rejected and the regression of y on x is significant.

(ii) Testing the Significance of the Difference Between an Observed Regression Coefficient and an Assumed Value ($H_0: \beta = \beta_0$):

The test statistic is $t = \frac{(b - \beta_0)}{s_b}$ with $(n-2)$ d.f.

On simplification, the test statistic becomes

$$t = \frac{(b - \beta_0) SS(x) \sqrt{n-2}}{\sqrt{SS(x)SS(y) - \{SP(xy)\}^2}}$$

Example 10.17:

Suppose on the basis of the data in example 10.16. We wish to test the hypothesis $H_0: \beta = 4.5$

$$\text{The test statistic is } t = \frac{(b - \beta_0) SS(x) \sqrt{n-2}}{\sqrt{SS(x)SS(y) - \{SP(xy)\}^2}}$$

$$= \frac{0.529 \times 171.67 \sqrt{10}}{\sqrt{158790.55}} = 0.721$$

which is smaller than the critical value at 5% level of significance.

Hence the null hypothesis may be accepted.

10.5. Tests of Population Variance :

- (i) Testing the significance of the difference between a population variance and an assumed value.

Let $(x_1, x_2, \dots, x_n)$ be a random sample from $N(\mu, \sigma^2)$ population on the basis of this sample we wish to test the hypothesis

$$H_0 : \sigma^2 = \sigma_0^2$$

For which the test statistic is a χ^2 (chi-square) variate with $(n-1)$ d.f.

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} \text{ with } (n-1) \text{ d.f.}$$

Where s^2 is the sample variance.

The test may be either one tailed or two tailed.

Example 10.17 :

A machine produces bolts. The distribution of bolts produced per minute follows a $N(\mu, \sigma^2)$ distribution. The following are the number of bolts produced in 10 randomly selected periods of one minute duration each :

10, 12, 13, 11, 13, 13, 12, 14, 10, 12

To test the hypothesis $H_0 : \sigma^2 = \sigma_0^2 = 1$,

We first compute, $s^2 = \frac{1}{9} \sum (x_i - \bar{x})^2 = 1.333$

Our test statistic is $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{9 \times 1.333}{1} = 11.997$

The 5% critical value of χ^2 with 9 d.f. is 16.91.

As the calculated value of χ^2 is less than the critical value there is no reason to reject the null hypothesis.

(ii) Testing the Significance of the Difference Between Two Population Variances :

Let $(x_{11}, x_{12}, \dots, x_{1n_1})$ and $(x_{21}, x_{22}, \dots, x_{2n_2})$ be two random samples of normal populations with means μ_1 and μ_2 respectively and variances σ_1^2 and σ_2^2 respectively.

We wish to test the hypothesis $H_0 : \sigma_1^2 = \sigma_2^2$

Our test statistic, the variance ratio follows Fisher's F distribution as defined below :

$$F = \frac{s_1^2}{s_2^2} \text{ with } (n_1 - 1) \text{ and } (n_2 - 1) \text{ d.f. if } s_1^2 > s_2^2$$

$$= \frac{s_2^2}{s_1^2} \text{ with } (n_2 - 1) \text{ and } (n_1 - 1) \text{ d.f. if } s_2^2 > s_1^2$$

The test may be one tailed or two-tailed.

Example 10.18 :

Let two random samples from a certain normal population give

$$n_1 = 4 \text{ and } n_2 = 6$$

$$s_1^2 = 0.087 \text{ and } s_2^2 = 0.34$$

Our test statistic is $F = \frac{s_2^2}{s_1^2} = \frac{0.34}{0.087} = 3.9$ with 5 and 3 d.f.

The critical value of $F_{5,3} (0.05) = 5.41$.

The test does not lead to reject the null hypothesis.

10.6. Test of Goodness of fit :

An important problem of statistical inference is to test the hypothesis that the given data have been obtained by random sampling from a specified population with definite value(s) for its parameter(s). The given data can be arranged in the form of a frequency distribution with observed frequencies for the various

classes. The corresponding expected (theoretical) frequencies are obtained on the basis of population distribution. Our aim is to examine the compatibility of the observed and the expected frequencies,

If $O_1, O_2, \dots, O_k$ are the observed frequencies of the k classes of the frequency distribution and the expected frequencies are $E_1, E_2, \dots, E_k$ respectively such that $\sum_{i=1}^k O_i = \sum_{i=1}^k E_i = N$.

The test statistic

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} = \sum_{i=1}^k \frac{O_i^2}{E_i} - N$$

is distributed as χ^2 (chi-square) distribution with $(k-1)$ d.f.

χ^2 Test of Goodness of Fit is a Large Sample Test Process and There are Certain Conditions for the Validity of the Test :

1. Sample observations should be independent.
2. Number of observations should be large ($N > 50$).
3. The expected frequency in each cell should be at least 5. If the expected frequency of any class is less than 5, the group is combined with its preceding or following group (whose expected frequency is at least 5) and for this the degrees of freedom of the χ^2 statistic will be reduced.

Example 10.20 :

The number of road accidents in different weekdays are given below:

Day:	Mon-day	Tues-day	Wednes-day	Thurs-day	Fri-day	Satur-day	Total
No. of accidents	14	18	12	11	15	14	84

We wish to test the hypothesis :

H_0 : Occurrence rate of accidents is same in all the weekdays.

Under the null hypothesis the expected number of accidents per day is $\frac{84}{6} = 14$, so the expected frequency in each day is 14. The test statistic is

$$\begin{aligned} \chi^2 &= \sum (O_i^2 / E_i) - N \quad \text{with } (6-1) = 5 \text{ d.f.} \\ &= \left(\frac{14^2}{14} + \frac{18^2}{14} + \frac{12^2}{14} + \frac{11^2}{14} + \frac{15^2}{14} + \frac{14^2}{14} \right) - 84 = 2.143. \end{aligned}$$

The 5% value of χ^2 with 5 d.f. is 11.07.

The test does not lead to rejection of the null hypothesis.

Example 10.21:

Five coins are tossed 1000 times and the number of flowers appeared are recorded as shown in the following frequency distribution:

No. of flowers (x)	0	1	2	3	4	5
Frequency (f)	38	144	342	287	164	25

We wish to test whether the data fits well to binomial distribution. Here mean $\bar{x} = \frac{1}{N} \sum f_i x_i = 2.47$

Mean of binomial distribution is $np = 2.47 \therefore p = \frac{2.47}{5} = 0.494$

$\therefore q = 1 - 0.494 = 0.506$ and $p/q = 0.976284584$

The fitted distribution is

$$P(x) = \binom{5}{x} (0.494)^x (0.506)^{5-x}; x = 0, 1, 2, \dots, 5$$

$$\therefore P(0) = \binom{5}{0} (0.494)^0 (0.506)^5 = 0.03317$$

In order to test the goodness of fit we compute the expected frequencies using the recurrence formula

$$P(x) = \frac{n-x+1}{x} \frac{p}{q} P(x-1)$$

The process of computing the expected frequencies is shown below:

Number of flower (x)	p/q	$P(x) = \frac{n-x+1}{x} P(x-1)$	$fe = N \times P(x)$
0	0.976284585	0.03317	33
1		0.161916789	161.9 ≈ 162
2		0.316153	316
3		0.308656	308.7 ≈ 309
4		0.150668	150.7 ≈ 151
5		0.029435	29
			1000

$$\text{Now, } \chi^2 = \left(\frac{38^2}{33} + \frac{144^2}{162} + \frac{342^2}{316} + \frac{287^2}{309} + \frac{164^2}{151} + \frac{25^2}{29} \right) - 1000 = 8.134$$

The critical value of χ^2 with 4 degrees of freedom at 5% level of significance is 9.488.

The above result indicates that the data fit well to the binomial distribution.

Example 10.21:

Fit a Poisson distribution to the following data and test the goodness of fit.

x :	0	1	2	3	4	5	6
f :	275	72	30	7	5	2	1

$$\text{Here } N = 392 \quad \text{Mean, } m = \frac{1}{N} \sum f_i x_i = 0.482$$

The fitted Poisson distribution is

$$P(x) = \frac{e^{-0.482} (0.482)^x}{x!}; \quad x = 0, 1, 2, \dots, 6$$

$$\text{Now, } p(0) = \frac{e^{-0.482} (0.482)^0}{0!} = e^{-0.482} = 0.617547061$$

The recurrence formula for Poisson distribution is $P(x) = \frac{m}{x} P(x-1)$

Expected frequencies are obtained using the recurrence formula as shown below :

x	$P(x) = \frac{0.482}{x} P(x-1)$	Expected frequency $f_e = N \times P(x)$	Observed frequency f_o
0	0.61754761	242.1 ≈ 242	275
1	0.2976579	116.6 ≈ 117	72
2	0.071736	28.1 ≈ 28	30
3	0.0115256	4.5 ≈ 5	7
4	0.003888	0.5 ≈ 1	5
5	0.0003748	0.1 ≈ 0	2
6	0.0000301	0.0 ≈ 0	1

$$\text{Test statistic, } \chi^2 = \sum \frac{(f_o - f_e)^2}{f_e} = 35.45 \quad \text{with } (7-1-3=3) \text{ d.f.}$$

Critical value of χ^2 with 3 d.f. at 1% level of significance is 11.345.

As the test statistic is much higher than the critical value, the data cannot be considered to fit with the Poisson distribution.

10.7. Test of Independence of Attributes :

Qualitative variables cannot be measured but individuals may be classified according to qualitative characteristics. For example, a group of students may be classified according to their merit, or according to their father's profession, social status etc. The qualitative characteristics are termed as *attributes*. A group of individuals may be classified according two attributes in a 2-way table. Such a two-way table arranged on the basis of two attributes is called *contingency*

table. An $m \times n$ contingency table is constituted with m classes of one attribute and n classes of another attribute. Such a table consists of m columns (the number of classes of one attribute) and n rows (the number of classes of the other attribute) as shown below :

Class of attribute B	Class of attribute A						Total
	a_1	a_2		a_i		a_m	
b_1	O_{11}	O_{21}		O_{i1}		O_{m1}	B_1
b_2	O_{12}	O_{22}		O_{i2}		O_{m2}	B_2
$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$	$\vdots$
b_j	O_{1j}	O_{2j}		O_{ij}		O_{mj}	B_j
$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$	$\vdots$
b_n	O_{1n}	O_{2n}		O_{in}		O_{mn}	B_n
Total	A_1	A_2		A_i		A_m	$N = \sum A_i = \sum B_j$

The above table is an $m \times n$ contingency table and contains $m \times n$ cells. The number of observations corresponding to the i th class of the attribute A and the j th class of the attribute B is O_{ij} which is the observed frequency of the ij -th cell. The table apparently looks like a correlation table. However, in correlation table, the variables are quantitative and in contingency table the variables are qualitative or categorical.

The relationship between the attributes in a contingency table is termed as association. If the association between the attributes is insignificant, the attributes are said to be independent. Independence of attributes in contingency tables are tested using χ^2 (chi-square) statistic and the test is known as χ^2 test of independence. The null hypothesis may be stated as

H_0 : There is no association between the attributes

or

H_0 : The attributes are independent

The test statistic is

$$\chi^2 = \sum_i \sum_j \left\{ \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \right\} = \sum_i \sum_j \frac{O_{ij}^2}{E_{ij}} - N$$

has χ^2 distribution with $(m-1)(n-1)$ d.f.

Here E_{ij} is the expected frequency of the ij -th cell, i.e. for the j th category of B in the i th category of A. E_{ij} is obtained as

$$E_{ij} = \frac{A_i \times B_j}{N}$$

where A_i is the total frequency for the i th category of A and B_j is the total frequency for the j th category of B. The test statistic is compared with the theoretical value of χ^2 with $(m-1)(n-1)$ degrees of freedom to comment on the acceptability of the null hypothesis.

Note :

1. If more than 20% cell frequencies in a contingency table are less than 5, χ^2 test may lead to misleading results.
2. A significant χ^2 leads to rejection of the null hypothesis but does not measure the degree of association or dependence of the attributes. A measure of dependence is given by coefficient of contingency.

Coefficient of Contingency :

The coefficient of contingency may be used as a measure of association of nominal data or data that may be placed in categories. The coefficient of contingency, denoted by C , is obtained by the following formula :

$$C = \sqrt{\frac{\chi^2}{N + \chi^2}}$$

The contingency coefficient, C , may be used without making any assumption about the nature of the distribution of the variables used. The coefficient C does not have 1 as an upper limit, the upper limit being related to the number of categories. For a table made up of equal number of columns and rows, k by k , the upper limit is $\sqrt{(k-1)/k}$: Thus for a 3×3 table the upper limit is $\sqrt{2/3} = 0.82$, for a 4×4 table, $\sqrt{3/4} = 0.87$ and so on. When the number of columns and rows differ, as in 3×4 , the upper limit follows that of the smaller number. The test of significance of C is equivalent to test the significance of χ^2 . If the χ^2 is significant, so is C .

Example 10.22:

A sample survey of BAU graduates employed in various organizations gave the follow results:

Type of organization	Utilization of educational qualification in present job			Total
	Fully utilized	Partially utilized	Very little or not all	
Education & research	112	19	6	137
Extension	88	27	1	116
Others	27	40	26	93
Total	227	86	33	346

We wish to examine the pattern of utilization of educational qualification in various types of organizations. The null hypothesis is

H_0 : There is no association between the utilization of education with the type of employer organization.

Here $N = 346$ and there are 3 classes of each attribute. Using the formula $E_{ij} = \frac{A_i \times B_j}{N}$ we obtain the expected cell frequencies as tabulated below:

Type of Employer	Expected frequencies			Total
	Fully	Partially	Very little or not all	
Education & research	90	34	13	137
Extension	76	29	11	116
Others	61	23	9	93
Total	227	86	33	346

$$\text{Test statistic } \chi^2 = \sum \frac{O_{ij}^2}{E_{ij}} - N$$

$$= \left\{ \frac{112^2}{90} + \frac{19^2}{34} + \frac{6^2}{13} + \frac{88^2}{76} + \frac{27^2}{29} + \frac{1^2}{11} + \frac{27^2}{61} + \frac{40^2}{23} + \frac{26^2}{9} \right\} - 346$$

$$= 90.52$$

The degrees of freedom is $(3-1) \times (3-1) = 4$

The critical value of χ^2 at 1% level of significance is 13.28. The computed value of χ^2 leads to rejection of the null hypothesis.

2 x 2 Contingency Table: If each of the attributes has two classes, the resulting contingency table is a 2×2 contingency table. In such case the test statistic can be directly obtained using the observed cell frequencies.

Suppose the following 2 x 2 contingency table is constituted with attributes A and B.

Classes of B	Classes of A		Total
	1	2	
1	a	b	a + b
2	c	d	c + d
Total	a + c	b + d	a + b + c + d = n

The test statistic is

$$\chi^2 = \frac{n(ad-bc)^2}{(a+b)(a+c)(b+d)(c+d)} \text{ with } (2-1) \times (2-1) = 1 \text{ d.f.}$$

Derivation of χ^2 Statistic for 2 x 2 Contingency Table :

Expected cell frequencies are obtained as

$$E(a) = \frac{(a+b)(a+c)}{n} \quad E(b) = \frac{(a+b)(b+d)}{n}$$

$$E(c) = \frac{(a+c)(c+d)}{n} \quad \text{and} \quad E(d) = \frac{(b+d)(c+d)}{n}$$

$$\begin{aligned} \text{Now } \frac{\{a - E(a)\}^2}{E(a)} &= \frac{\left\{a - \frac{(a+b)(a+c)}{n}\right\}^2}{\frac{(a+b)(a+c)}{n}} \\ &= \frac{\left\{\frac{(a^2 + ab + ac + ad) - (a^2 + ac + ab + bc)}{(a+b+c+d)}\right\}^2}{\frac{(a+b)(a+c)}{n}} \\ &= \frac{(ad-bc)^2}{(a+b+c+d)^2} \cdot \frac{n^2}{(a+b)(a+c)} = \frac{(ad-bc)^2}{n(a+b)(a+c)} \end{aligned}$$

Similarly it can be shown that

$$\frac{\{b - E(b)\}^2}{E(b)} = \frac{(ad-bc)^2}{n(a+b)(b+d)} \quad \frac{\{c - E(c)\}^2}{E(c)} = \frac{(ad-bc)^2}{n(a+c)(c+d)} \text{ and}$$

$$\frac{\{d - E(d)\}^2}{E(d)} = \frac{(ad-bc)^2}{n(b+d)(c+d)}$$

By definition

$$\begin{aligned} \chi^2 &= \frac{\{a - E(a)\}^2}{E(a)} + \frac{\{b - E(b)\}^2}{E(b)} + \frac{\{c - E(c)\}^2}{E(c)} + \frac{\{d - E(d)\}^2}{E(d)} \\ &= \frac{(ad-bc)^2}{n(a+b)(a+c)} + \frac{(ad-bc)^2}{n(a+b)(b+d)} + \frac{(ad-bc)^2}{n(a+c)(c+d)} + \frac{(ad-bc)^2}{n(b+d)(c+d)} \\ &= \frac{(ad-bc)^2}{n} \left\{ \frac{1}{(a+b)(a+c)} + \frac{1}{(a+b)(b+d)} + \frac{1}{(a+c)(c+d)} + \frac{1}{(b+d)(c+d)} \right\} \\ &= \frac{(ad-bc)^2}{n} \left\{ \frac{(b+d)(c+d) + (a+c)(c+d) + (a+b)(b+d) + (a+b)(a+c)}{(a+b)(a+c)(b+d)(c+d)} \right\} \\ &= \frac{(ad-bc)^2}{n} \left\{ \frac{bc + bd + cd + d^2 + ac + ad + c^2 + cd + ab + ad + b^2 + bd + a^2 + ac + ab + bc}{(a+b)(a+c)(b+d)(c+d)} \right\} \\ &= \frac{(ad-bc)^2}{n} \left\{ \frac{a^2 + b^2 + c^2 + d^2 + 2ab + 2ac + 2ad + 2bc + 2bd + 2cd}{(a+b)(a+c)(b+d)(c+d)} \right\} \\ &= \frac{(ad-bc)^2}{n} \left\{ \frac{(a+b+c+d)^2}{(a+b)(a+c)(b+d)(c+d)} \right\} = \frac{(ad-bc)^2}{n} \cdot \frac{n^2}{(a+b)(a+c)(b+d)(c+d)} \\ &= \frac{n(ad-bc)^2}{(a+b)(a+c)(b+d)(c+d)} \end{aligned}$$

Example 10.23:

The following are the results of an investigation to examine the effectiveness of a certain vaccine used to prevent a specific disease of domestic animals:

	Vaccinated	Not vaccinated	Total
Diseased	12	16	28
Non-diseased	26	6	32
Total	38	22	60

We wish to test the hypothesis

H_0 : The vaccine is not effective to prevent the disease

$$\text{Test statistic } \chi^2 = \frac{n(ad - bc)^2}{(a+b)(a+c)(b+d)(c+d)}$$

$$= \frac{60(12 \times 6 - 26 \times 16)^2}{38 \times 22 \times 28 \times 32} = 9.479$$

The 5% and 1% critical value of χ^2 with 1 d.f. are 3.841 and 6.635 respectively. Hence the null hypothesis is rejected at 1% level of significance and the vaccination proves to be very effective.

Yate's Correction for 2 x 2 Contingency Table:

χ^2 distribution is a continuous distribution, but the distribution of discrete numbers is naturally discrete. However, for large sample ($n \geq 50$) the test statistic based on a 2 x 2 contingency table may be considered to be approximately distributed as χ^2 distribution. But small cell frequency (< 5) affects the continuity of χ^2 distribution. As a remedy, F. Yates (1934) suggests a correction for continuity, which is known as Yate's correction. However, it is advisable to use Yate's

correction in the following situation even if no cell frequency is less than 5.

Let the cell frequencies of a 2 x 2 contingency table be a, b, c and d.

- If $ad > bc$, $\frac{1}{2}$ be deducted from each of a & d and $\frac{1}{2}$ be added to each of b and c.
- If $ad < bc$, $\frac{1}{2}$ be added with each of a & d and $\frac{1}{2}$ be deducted from each of b and c.

After Yate's correction the formula for χ^2 statistic becomes

$$\chi^2 = \frac{n \left\{ |ad - bc| - \frac{n}{2} \right\}^2}{(a+b)(a+c)(b+d)(c+d)}$$

Example 10.24:

Preventives were applied for Ranikhet disease in a small poultry farm and after one month the following situation was found.

Situation	Application of preventive		Total
	Applied	Not applied	
Number died	3	35	38
Number alive	47	15	62
Total	50	50	100

We wish to examine the effect of preventive application for Ranikhet. The null hypothesis is

H_0 : The medicine has no effect in preventing Ranikhet.

Here one cell frequency is less than 5; so we need to use Yates' correction for χ^2 -statistic.

We have $n=100$, $ad = 45$ and $bc = 1645$.

$$\therefore \chi^2 = \frac{n \left\{ |ad - bc| - \frac{n}{2} \right\}^2}{(a+b)(a+c)(b+d)(c+d)}$$

$$= \frac{100 \left\{ |45 - 1645| - \frac{100}{2} \right\}^2}{50 \times 50 \times 38 \times 62} = 46.22$$

The 1% critical value of χ^2 with 1 d.f. is 6.635. Our result indicates that the application of medicine is highly effective in preventing Ranikhet disease.

10.7.1. Fishers' Exact Probability Test:

If one or more cell frequencies in a 2×2 contingency table is less than 5 and the sample is large enough ($n \geq 50$) χ^2 test can be applied using Yate's correction discussed above, but if the sample is not large enough ($n < 50$), χ^2 test cannot be used. In such situation test of independence is done using Fisher's exact probability test as discussed below:

Let a, b, c and d be the cell frequencies of a 2×2 contingency table. The body of this contingency table may be arranged in.

$$\frac{n!}{a!b!c!d!} \text{ ways, where } n = a + b + c + d.$$

And under null hypothesis (assuming independence of attributes) the total number of ways in which marginal totals can be obtained is

$$\frac{n!}{(a+c)!(b+d)!} \times \frac{n!}{(a+b)!(c+d)!}$$

The probability of a is obtained as,

$$P(a) = \frac{(a+c)!(b+d)!(a+b)!(c+d)!}{n!a!b!c!d!}$$

The null hypothesis is tested using

$$P = \sum_a p(a) \text{ for one tailed test.}$$

For two-tailed test $2P$ is used for the decision.

Example 10.25:

Suppose a sample consists of 24 animals of which 12 are given a certain preventive medicine and the other 12 are not give any preventive; let the results of the trial be tabulated as -

	Preventive applied	Preventive not applied	Total
Animals died	2	8	10
Animals alive	10	4	14
Total	12	12	24

Here the smallest frequency is $a = 2$.

So for convenience, we will compute the probability of $a = 0, 1$, and 2.

Keeping the marginal totals unchanged,

if $a = 0$; b, c and d will be 12, 10 and 2 respectively;

if $a = 1$; b, c and d will be 11, 9 and 3 respectively and

if $a = 2$; b, c and d will be 10, 8 and 4 respectively.

$$\text{Now } P(a=0) = \frac{(12)!(12)!(10)!(14)!}{(0)!(12)!(10)!(12)!(24)!}$$

$$P(a=1) = \frac{(12)!(12)!(10)!(14)!}{(1)!(11)!(9)!(3)!(24)!}$$

$$P(a=2) = \frac{(12)!(12)!(10)!(14)!}{(2)!(10)!(8)!(4)!(24)!}$$

$$P = P(a=0) + P(a=1) + P(a=2)$$

$$= \frac{(12)!(12)!(10)!(14)!}{(0)!(12)!(10)!(12)!(24)!} + \frac{(12)!(12)!(10)!(14)!}{(1)!(11)!(9)!(3)!(24)!}$$

$$+ \frac{(12)!(12)!(10)!(14)!}{(2)!(10)!(8)!(4)!(24)!}$$

$$= \frac{(12)!(14)!}{(24)!(2)!} \left\{ 1 + \frac{12 \times 10}{1 \times 3} + \frac{12 \times 11 \times 10 \times 9}{1 \times 12 \times 3 \times 4} \right\} = 0.018$$

$$2P = 2 \times 0.018 = 0.036$$

Here $2P < 0.05$

So the null hypothesis is rejected.

NONPARAMETRIC STATISTICAL TESTS

Nonparametric statistical tests (often called distribution free tests) are those which do not require to specify the condition of the parameters of the population from which the sample has been drawn. No assumption about the precise form of the population is made. Sometimes certain assumptions are made such that a distribution is continuous or that the sampled populations have identical shapes or distributions are symmetrical about the same point. However, the assumptions are never so elaborate as to imply a completely specified population distribution, as in the case with the normal curve.

Some of the major points regarding advantages and disadvantages of nonparametric statistical tests are the following :

1. Simplicity of Derivation : The derivation of classical tests requires a level of competence in mathematics far above that attained by typical research workers, whereas most distribution free statistics can be derived using simple formulae.

2. Ease of Application : Frequently, the only mathematical operation needed for nonparametric tests are ranking, counting, adding and subtracting.

3. Speed of Application : When sample size is small or moderate nonparametric methods are generally faster than parametric techniques.

4. Scope of Application : Because they are based on fewer and less elaborate assumptions than the classical tests, nonparametric tests can be properly applied to much larger class of population.

5. Susceptibility to Violation of Assumptions : Since the assumptions are fewer and less elaborate with nonparametric tests, they are less susceptible to violations. These violations are easier to